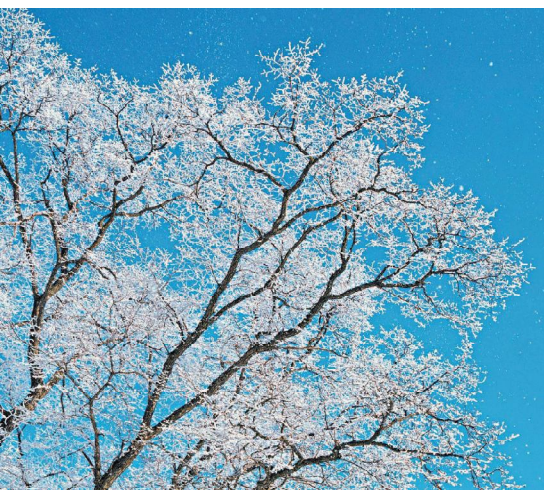




14

PARTIAL DERIVATIVES

OVERVIEW Many functions depend on more than one independent variable. For instance, the volume of a right circular cylinder is a function $V = \pi r^2 h$ of its radius and its height, so it is a function $V(r, h)$ of two variables r and h . In this chapter we extend the basic ideas of single variable calculus to functions of several variables. Their derivatives are more varied and interesting because of the different ways the variables can interact. The applications of these derivatives are also more varied than for single-variable calculus, and in the next chapter we will see that the same is true for integrals involving several variables.

14.1 | Functions of Several Variables

In this section we define functions of more than one independent variable and discuss ways to graph them.

Real-valued functions of several independent real variables are defined similarly to functions in the single-variable case. Points in the domain are ordered pairs (triples, quadruples, n -tuples) of real numbers, and values in the range are real numbers as we have worked with all along.

DEFINITIONS Suppose D is a set of n -tuples of real numbers $(x_1, x_2, \dots, x_n)$. A **real-valued function** f on D is a rule that assigns a unique (single) real number

$$w = f(x_1, x_2, \dots, x_n)$$

to each element in D . The set D is the function's **domain**. The set of w -values taken on by f is the function's **range**. The symbol w is the **dependent variable** of f , and f is said to be a function of the n **independent variables** x_1 to x_n . We also call the x_j 's the function's **input variables** and call w the function's **output variable**.

If f is a function of two independent variables, we usually call the independent variables x and y and the dependent variable z , and we picture the domain of f as a region in the xy -plane (Figure 14.1). If f is a function of three independent variables, we call the independent variables x , y , and z and the dependent variable w , and we picture the domain as a region in space.

In applications, we tend to use letters that remind us of what the variables stand for. To say that the volume of a right circular cylinder is a function of its radius and height, we might write $V = f(r, h)$. To be more specific, we might replace the notation $f(r, h)$ by the formula that calculates the value of V from the values of r and h , and write $V = \pi r^2 h$. In either case, r and h would be the independent variables and V the dependent variable of the function.

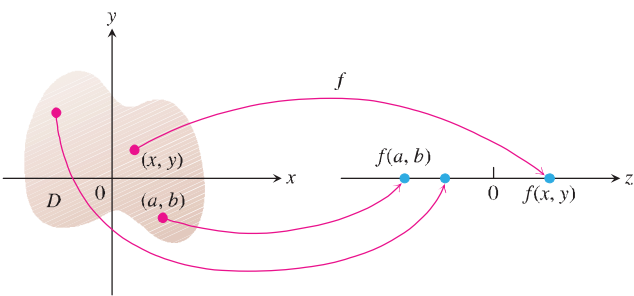


FIGURE 14.1 An arrow diagram for the function $z = f(x, y)$.

As usual, we evaluate functions defined by formulas by substituting the values of the independent variables in the formula and calculating the corresponding value of the dependent variable. For example, the value of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at the point $(3, 0, 4)$ is

$$f(3, 0, 4) = \sqrt{(3)^2 + (0)^2 + (4)^2} = \sqrt{25} = 5.$$

Domains and Ranges

In defining a function of more than one variable, we follow the usual practice of excluding inputs that lead to complex numbers or division by zero. If $f(x, y) = \sqrt{y - x^2}$, y cannot be less than x^2 . If $f(x, y) = 1/(xy)$, xy cannot be zero. The domain of a function is assumed to be the largest set for which the defining rule generates real numbers, unless the domain is otherwise specified explicitly. The range consists of the set of output values for the dependent variable.

EXAMPLE 1 (a) These are functions of two variables. Note the restrictions that may apply to their domains in order to obtain a real value for the dependent variable z .

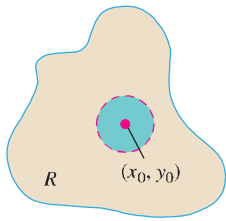
Function	Domain	Range
$z = \sqrt{y - x^2}$	$y \geq x^2$	$[0, \infty)$
$z = \frac{1}{xy}$	$xy \neq 0$	$(-\infty, 0) \cup (0, \infty)$
$z = \sin xy$	Entire plane	$[-1, 1]$

(b) These are functions of three variables with restrictions on some of their domains.

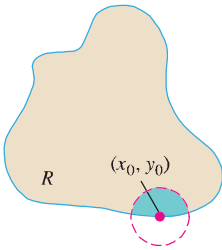
Function	Domain	Range
$w = \sqrt{x^2 + y^2 + z^2}$	Entire space	$[0, \infty)$
$w = \frac{1}{x^2 + y^2 + z^2}$	$(x, y, z) \neq (0, 0, 0)$	$(0, \infty)$
$w = xy \ln z$	Half-space $z > 0$	$(-\infty, \infty)$

Functions of Two Variables

Regions in the plane can have interior points and boundary points just like intervals on the real line. Closed intervals $[a, b]$ include their boundary points, open intervals (a, b) don't include their boundary points, and intervals such as $[a, b)$ are neither open nor closed.



(a) Interior point



(b) Boundary point

FIGURE 14.2 Interior points and boundary points of a plane region R . An interior point is necessarily a point of R . A boundary point of R need not belong to R .

DEFINITIONS A point (x_0, y_0) in a region (set) R in the xy -plane is an **interior point** of R if it is the center of a disk of positive radius that lies entirely in R (Figure 14.2). A point (x_0, y_0) is a **boundary point** of R if every disk centered at (x_0, y_0) contains points that lie outside of R as well as points that lie in R . (The boundary point itself need not belong to R .)

The interior points of a region, as a set, make up the **interior** of the region. The region's boundary points make up its **boundary**. A region is **open** if it consists entirely of interior points. A region is **closed** if it contains all its boundary points (Figure 14.3).

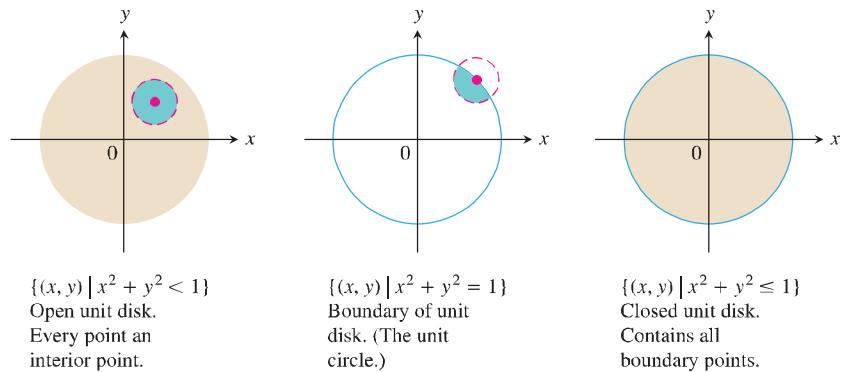


FIGURE 14.3 Interior points and boundary points of the unit disk in the plane.

As with a half-open interval of real numbers $[a, b)$, some regions in the plane are neither open nor closed. If you start with the open disk in Figure 14.3 and add to it some of but not all its boundary points, the resulting set is neither open nor closed. The boundary points that *are* there keep the set from being open. The absence of the remaining boundary points keeps the set from being closed.

DEFINITIONS A region in the plane is **bounded** if it lies inside a disk of fixed radius. A region is **unbounded** if it is not bounded.

Examples of *bounded* sets in the plane include line segments, triangles, interiors of triangles, rectangles, circles, and disks. Examples of *unbounded* sets in the plane include lines, coordinate axes, the graphs of functions defined on infinite intervals, quadrants, half-planes, and the plane itself.

EXAMPLE 2 Describe the domain of the function $f(x, y) = \sqrt{y - x^2}$.

Solution Since f is defined only where $y - x^2 \geq 0$, the domain is the closed, unbounded region shown in Figure 14.4. The parabola $y = x^2$ is the boundary of the domain. The points above the parabola make up the domain's interior.

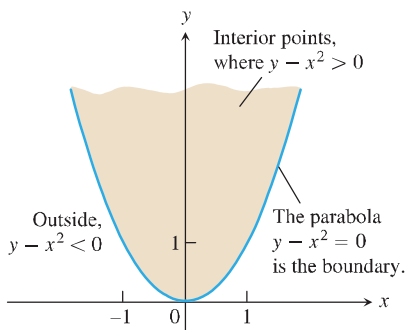


FIGURE 14.4 The domain of $f(x, y)$ in Example 2 consists of the shaded region and its bounding parabola.

Graphs, Level Curves, and Contours of Functions of Two Variables

There are two standard ways to picture the values of a function $f(x, y)$. One is to draw and label curves in the domain on which f has a constant value. The other is to sketch the surface $z = f(x, y)$ in space.

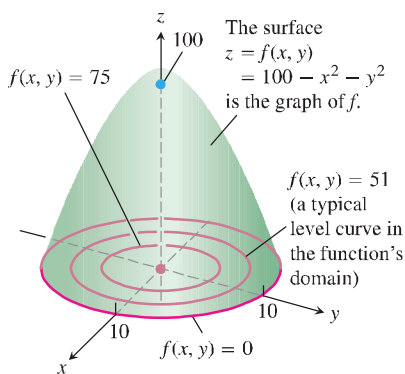


FIGURE 14.5 The graph and selected level curves of the function $f(x, y)$ in Example 3.

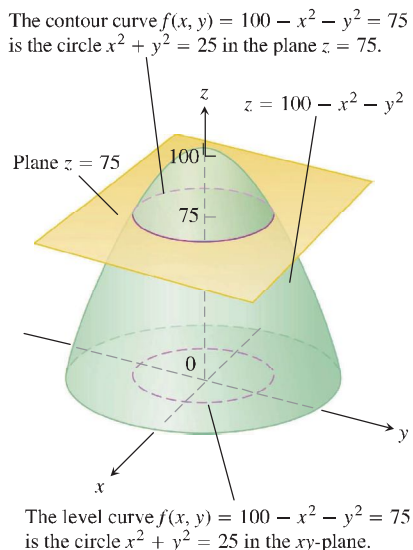


FIGURE 14.6 A plane $z = c$ parallel to the xy -plane intersecting a surface $z = f(x, y)$ produces a contour curve.

DEFINITIONS The set of points in the plane where a function $f(x, y)$ has a constant value $f(x, y) = c$ is called a **level curve** of f . The set of all points $(x, y, f(x, y))$ in space, for (x, y) in the domain of f , is called the **graph** of f . The graph of f is also called the **surface** $z = f(x, y)$.

EXAMPLE 3 Graph $f(x, y) = 100 - x^2 - y^2$ and plot the level curves $f(x, y) = 0$, $f(x, y) = 51$, and $f(x, y) = 75$ in the domain of f in the plane.

Solution The domain of f is the entire xy -plane, and the range of f is the set of real numbers less than or equal to 100. The graph is the paraboloid $z = 100 - x^2 - y^2$, the positive portion of which is shown in Figure 14.5.

The level curve $f(x, y) = 0$ is the set of points in the xy -plane at which

$$f(x, y) = 100 - x^2 - y^2 = 0, \quad \text{or} \quad x^2 + y^2 = 100,$$

which is the circle of radius 10 centered at the origin. Similarly, the level curves $f(x, y) = 51$ and $f(x, y) = 75$ (Figure 14.5) are the circles

$$f(x, y) = 100 - x^2 - y^2 = 51, \quad \text{or} \quad x^2 + y^2 = 49$$

$$f(x, y) = 100 - x^2 - y^2 = 75, \quad \text{or} \quad x^2 + y^2 = 25.$$

The level curve $f(x, y) = 100$ consists of the origin alone. (It is still a level curve.)

If $x^2 + y^2 > 100$, then the values of $f(x, y)$ are negative. For example, the circle $x^2 + y^2 = 144$, which is the circle centered at the origin with radius 12, gives the constant value $f(x, y) = -44$ and is a level curve of f . ■

The curve in space in which the plane $z = c$ cuts a surface $z = f(x, y)$ is made up of the points that represent the function value $f(x, y) = c$. It is called the **contour curve** $f(x, y) = c$ to distinguish it from the level curve $f(x, y) = c$ in the domain of f . Figure 14.6 shows the contour curve $f(x, y) = 75$ on the surface $z = 100 - x^2 - y^2$ defined by the function $f(x, y) = 100 - x^2 - y^2$. The contour curve lies directly above the circle $x^2 + y^2 = 25$, which is the level curve $f(x, y) = 75$ in the function's domain.

Not everyone makes this distinction, however, and you may wish to call both kinds of curves by a single name and rely on context to convey which one you have in mind. On most maps, for example, the curves that represent constant elevation (height above sea level) are called contours, not level curves (Figure 14.7).

Functions of Three Variables

In the plane, the points where a function of two independent variables has a constant value $f(x, y) = c$ make a curve in the function's domain. In space, the points where a function of three independent variables has a constant value $f(x, y, z) = c$ make a surface in the function's domain.

DEFINITION The set of points (x, y, z) in space where a function of three independent variables has a constant value $f(x, y, z) = c$ is called a **level surface** of f .

Since the graphs of functions of three variables consist of points $(x, y, z, f(x, y, z))$ lying in a four-dimensional space, we cannot sketch them effectively in our three-dimensional frame of reference. We can see how the function behaves, however, by looking at its three-dimensional level surfaces.

EXAMPLE 4 Describe the level surfaces of the function

$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}.$$

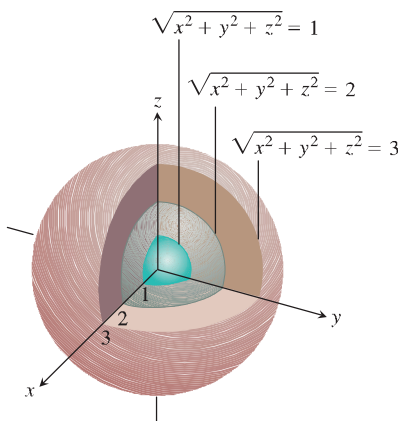


FIGURE 14.8 The level surfaces of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ are concentric spheres (Example 4).

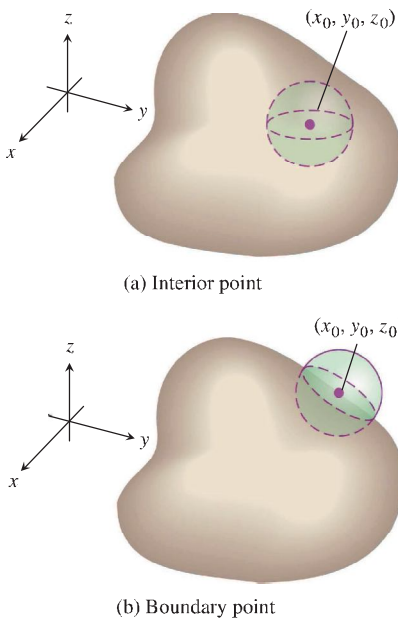


FIGURE 14.9 Interior points and boundary points of a region in space. As with regions in the plane, a boundary point need not belong to the space region R .

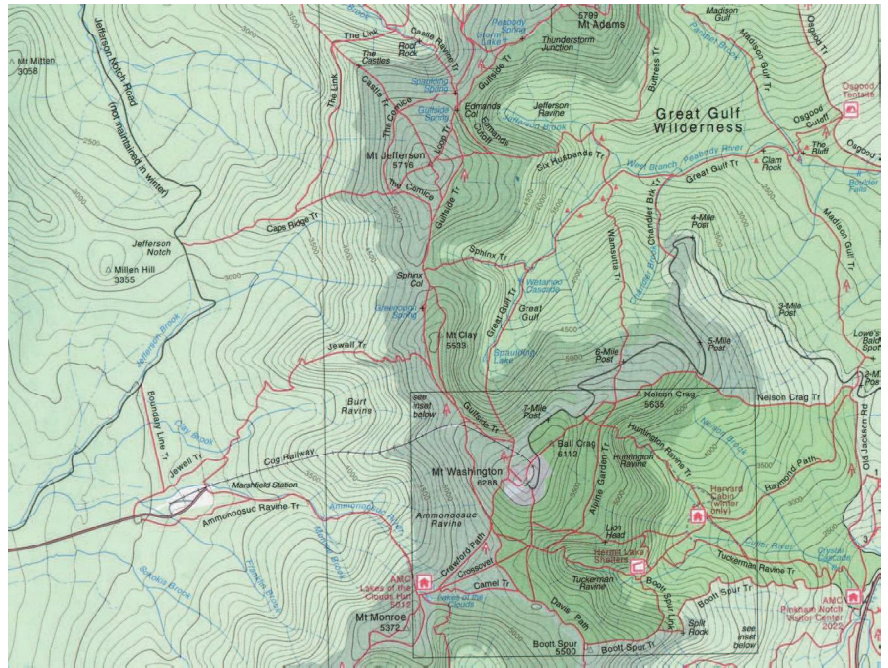


FIGURE 14.7 Contours on Mt. Washington in New Hampshire. (Reproduced by permission from the Appalachian Mountain Club.)

Solution The value of f is the distance from the origin to the point (x, y, z) . Each level surface $\sqrt{x^2 + y^2 + z^2} = c$, $c > 0$, is a sphere of radius c centered at the origin. Figure 14.8 shows a cutaway view of three of these spheres. The level surface $\sqrt{x^2 + y^2 + z^2} = 0$ consists of the origin alone.

We are not graphing the function here; we are looking at level surfaces in the function's domain. The level surfaces show how the function's values change as we move through its domain. If we remain on a sphere of radius c centered at the origin, the function maintains a constant value, namely c . If we move from a point on one sphere to a point on another, the function's value changes. It increases if we move away from the origin and decreases if we move toward the origin. The way the values change depends on the direction we take. The dependence of change on direction is important. We return to it in Section 14.5.

The definitions of interior, boundary, open, closed, bounded, and unbounded for regions in space are similar to those for regions in the plane. To accommodate the extra dimension, we use solid balls of positive radius instead of disks.

DEFINITIONS A point (x_0, y_0, z_0) in a region R in space is an **interior point** of R if it is the center of a solid ball that lies entirely in R (Figure 14.9a). A point (x_0, y_0, z_0) is a **boundary point** of R if every solid ball centered at (x_0, y_0, z_0) contains points that lie outside of R as well as points that lie inside R (Figure 14.9b). The **interior** of R is the set of interior points of R . The **boundary** of R is the set of boundary points of R .

A region is **open** if it consists entirely of interior points. A region is **closed** if it contains its entire boundary.

Examples of *open* sets in space include the interior of a sphere, the open half-space $z > 0$, the first octant (where x , y , and z are all positive), and space itself. Examples of *closed* sets in space include lines, planes, and the closed half-space $z \geq 0$. A solid sphere

with part of its boundary removed or a solid cube with a missing face, edge, or corner point is *neither open nor closed*.

Functions of more than three independent variables are also important. For example, the temperature on a surface in space may depend not only on the location of the point $P(x, y, z)$ on the surface but also on the time t when it is visited, so we would write $T = f(x, y, z, t)$.

Computer Graphing

Three-dimensional graphing programs for computers and calculators make it possible to graph functions of two variables with only a few keystrokes. We can often get information more quickly from a graph than from a formula.

EXAMPLE 5 The temperature w beneath the Earth's surface is a function of the depth x beneath the surface and the time t of the year. If we measure x in feet and t as the number of days elapsed from the expected date of the yearly highest surface temperature, we can model the variation in temperature with the function

$$w = \cos(1.7 \times 10^{-2}t - 0.2x)e^{-0.2x}.$$

(The temperature at 0 ft is scaled to vary from +1 to -1, so that the variation at x feet can be interpreted as a fraction of the variation at the surface.)

Figure 14.10 shows a graph of the function. At a depth of 15 ft, the variation (change in vertical amplitude in the figure) is about 5% of the surface variation. At 25 ft, there is almost no variation during the year.

The graph also shows that the temperature 15 ft below the surface is about half a year out of phase with the surface temperature. When the temperature is lowest on the surface (late January, say), it is at its highest 15 ft below. Fifteen feet below the ground, the seasons are reversed.

Figure 14.11 shows computer-generated graphs of a number of functions of two variables together with their level curves.

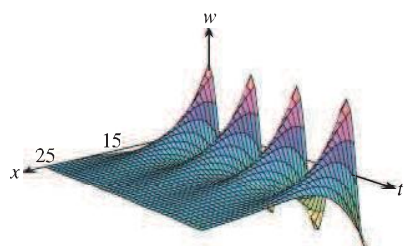


FIGURE 14.10 This graph shows the seasonal variation of the temperature below ground as a fraction of surface temperature (Example 5).

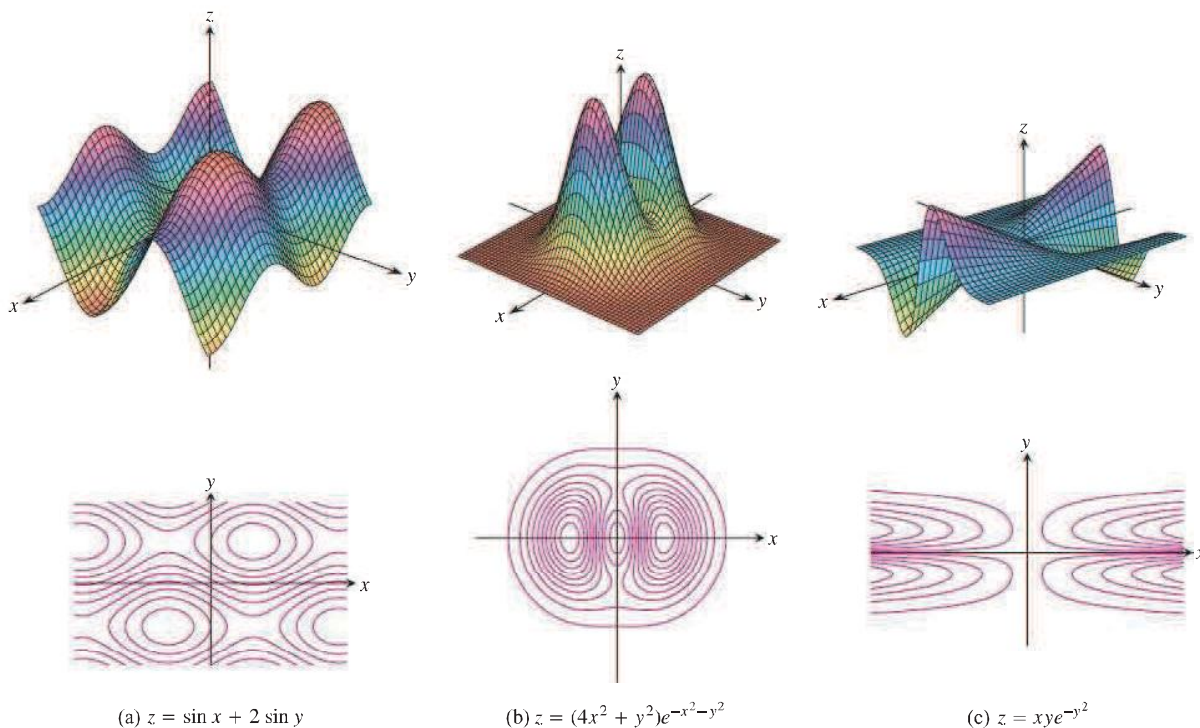


FIGURE 14.11 Computer-generated graphs and level curves of typical functions of two variables.

Exercises 14.1

Domain, Range, and Level Curves

In Exercises 1–4, find the specific function values.

1. $f(x, y) = x^2 + xy^3$
 - a. $f(0, 0)$
 - b. $f(-1, 1)$
 - c. $f(2, 3)$
 - d. $f(-3, -2)$
2. $f(x, y) = \sin(xy)$
 - a. $f\left(2, \frac{\pi}{6}\right)$
 - b. $f\left(-3, \frac{\pi}{12}\right)$
 - c. $f\left(\pi, \frac{1}{4}\right)$
 - d. $f\left(-\frac{\pi}{2}, -7\right)$
3. $f(x, y, z) = \frac{x - y}{y^2 + z^2}$
 - a. $f(3, -1, 2)$
 - b. $f\left(1, \frac{1}{2}, -\frac{1}{4}\right)$
 - c. $f\left(0, -\frac{1}{3}, 0\right)$
 - d. $f(2, 2, 100)$
4. $f(x, y, z) = \sqrt{49 - x^2 - y^2 - z^2}$
 - a. $f(0, 0, 0)$
 - b. $f(2, -3, 6)$
 - c. $f(-1, 2, 3)$
 - d. $f\left(\frac{4}{\sqrt{2}}, \frac{5}{\sqrt{2}}, \frac{6}{\sqrt{2}}\right)$

In Exercises 5–12, find and sketch the domain for each function.

5. $f(x, y) = \sqrt{y - x - 2}$
6. $f(x, y) = \ln(x^2 + y^2 - 4)$
7. $f(x, y) = \frac{(x - 1)(y + 2)}{(y - x)(y - x^3)}$
8. $f(x, y) = \frac{\sin(xy)}{x^2 + y^2 - 25}$
9. $f(x, y) = \cos^{-1}(y - x^2)$
10. $f(x, y) = \ln(xy + x - y - 1)$
11. $f(x, y) = \sqrt{(x^2 - 4)(y^2 - 9)}$
12. $f(x, y) = \frac{1}{\ln(4 - x^2 - y^2)}$

In Exercises 13–16, find and sketch the level curves $f(x, y) = c$ on the same set of coordinate axes for the given values of c . We refer to these level curves as a contour map.

13. $f(x, y) = x + y - 1$, $c = -3, -2, -1, 0, 1, 2, 3$
14. $f(x, y) = x^2 + y^2$, $c = 0, 1, 4, 9, 16, 25$
15. $f(x, y) = xy$, $c = -9, -4, -1, 0, 1, 4, 9$
16. $f(x, y) = \sqrt{25 - x^2 - y^2}$, $c = 0, 1, 2, 3, 4$

In Exercises 17–30, (a) find the function's domain, (b) find the function's range, (c) describe the function's level curves, (d) find the boundary of the function's domain, (e) determine if the domain is an open region, a closed region, or neither, and (f) decide if the domain is bounded or unbounded.

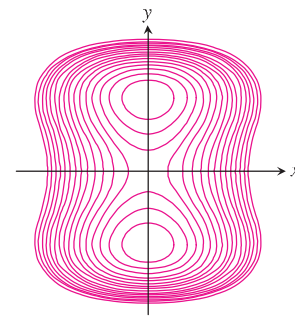
17. $f(x, y) = y - x$
18. $f(x, y) = \sqrt{y - x}$
19. $f(x, y) = 4x^2 + 9y^2$
20. $f(x, y) = x^2 - y^2$

21. $f(x, y) = xy$
22. $f(x, y) = y/x^2$
23. $f(x, y) = \frac{1}{\sqrt{16 - x^2 - y^2}}$
24. $f(x, y) = \sqrt{9 - x^2 - y^2}$
25. $f(x, y) = \ln(x^2 + y^2)$
26. $f(x, y) = e^{-(x^2 + y^2)}$
27. $f(x, y) = \sin^{-1}(y - x)$
28. $f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$
29. $f(x, y) = \ln(x^2 + y^2 - 1)$
30. $f(x, y) = \ln(9 - x^2 - y^2)$

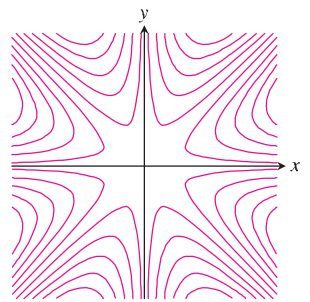
Matching Surfaces with Level Curves

Exercises 31–36 show level curves for the functions graphed in (a)–(f) on the following page. Match each set of curves with the appropriate function.

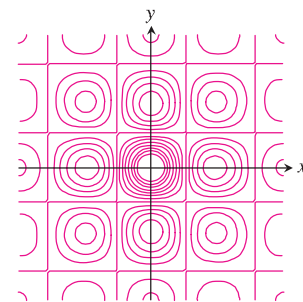
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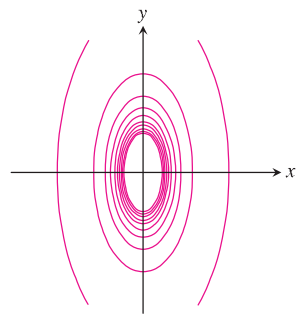
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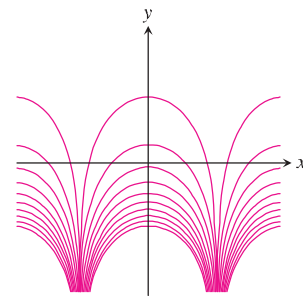
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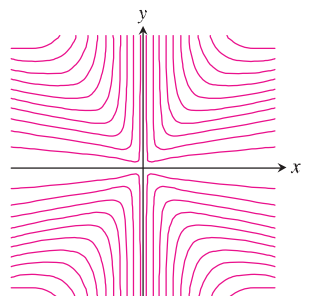
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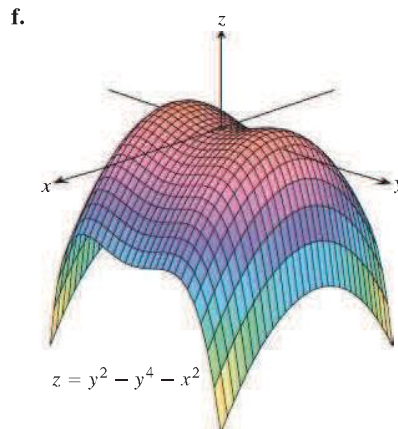
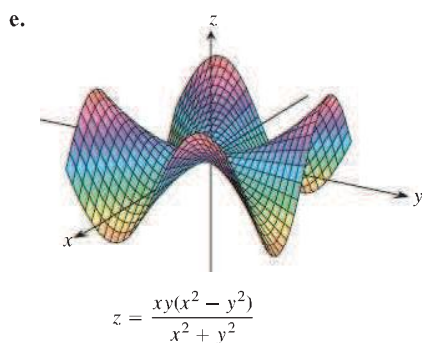
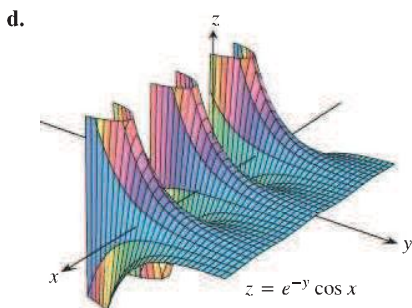
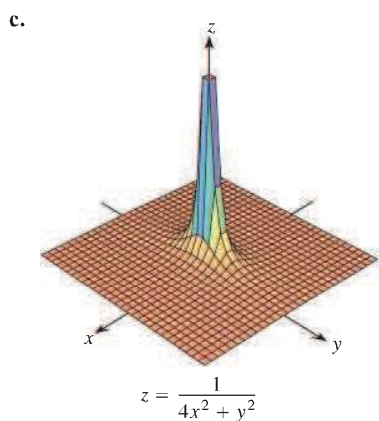
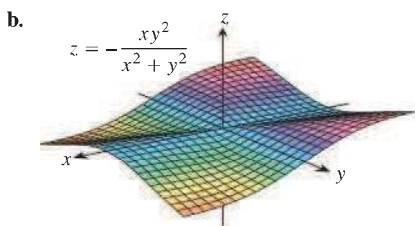
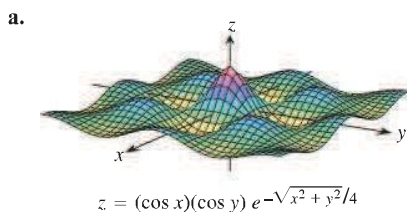


35.



36.





Functions of Two Variables

Display the values of the functions in Exercises 37–48 in two ways: (a) by sketching the surface $z = f(x, y)$ and (b) by drawing an assortment of level curves in the function's domain. Label each level curve with its function value.

- | | |
|--------------------------------------|--------------------------------------|
| 37. $f(x, y) = y^2$ | 38. $f(x, y) = \sqrt{x}$ |
| 39. $f(x, y) = x^2 + y^2$ | 40. $f(x, y) = \sqrt{x^2 + y^2}$ |
| 41. $f(x, y) = x^2 - y$ | 42. $f(x, y) = 4 - x^2 - y^2$ |
| 43. $f(x, y) = 4x^2 + y^2$ | 44. $f(x, y) = 6 - 2x - 3y$ |
| 45. $f(x, y) = 1 - y $ | 46. $f(x, y) = 1 - x - y $ |
| 47. $f(x, y) = \sqrt{x^2 + y^2 + 4}$ | 48. $f(x, y) = \sqrt{x^2 + y^2 - 4}$ |

Finding Level Curves

In Exercises 49–52, find an equation for and sketch the graph of the level curve of the function $f(x, y)$ that passes through the given point.

49. $f(x, y) = 16 - x^2 - y^2$, $(2\sqrt{2}, \sqrt{2})$
50. $f(x, y) = \sqrt{x^2 - 1}$, $(1, 0)$
51. $f(x, y) = \sqrt{x + y^2 - 3}$, $(3, -1)$
52. $f(x, y) = \frac{2y - x}{x + y + 1}$, $(-1, 1)$

Sketching Level Surfaces

In Exercises 53–60, sketch a typical level surface for the function.

53. $f(x, y, z) = x^2 + y^2 + z^2$
54. $f(x, y, z) = \ln(x^2 + y^2 + z^2)$
55. $f(x, y, z) = x + z$
56. $f(x, y, z) = z$
57. $f(x, y, z) = x^2 + y^2$
58. $f(x, y, z) = y^2 + z^2$
59. $f(x, y, z) = z - x^2 - y^2$
60. $f(x, y, z) = (x^2/25) + (y^2/16) + (z^2/9)$

Finding Level Surfaces

In Exercises 61–64, find an equation for the level surface of the function through the given point.

61. $f(x, y, z) = \sqrt{x - y} - \ln z$, $(3, -1, 1)$
62. $f(x, y, z) = \ln(x^2 + y + z^2)$, $(-1, 2, 1)$